Deciding whether to undergo a mastectomy is a difficult and potentially fatal choice. Beyond the discomforts, dangers and financial costs of the operation and its follow-up treatments, the decision can have far-reaching effects on one's health, body image, emotional well-being and relationships. Jolie's choice, and the courage she showed in going public with it, caused a great stir and won her international acclaim and admiration. In particular, many hoped that the publicity would increase awareness of genetic medicine and its potential benefits.

From a historical perspective, it is interesting to note the critical role algorithms played in this case. When Jolie had to take such an important decision about her life, she did not climb a mountaintop overlooking the ocean, watch the sun set into the waves and attempt to connect to her innermost feelings. Instead, she preferred to listen to her genes, whose voice manifested not in feelings but in numbers. Jolie felt no pain or discomfort whatsoever. Her feelings told her: 'Relax, everything is perfectly fine.' But the computer algorithms used by her doctors told a different story: 'You don't feel anything is wrong, but there is a time bomb ticking in your DNA. Do something about it – now!'

Of course, Jolie's emotions and unique personality played a key part too. If another woman with a different personality had discovered she was carrying the same genetic mutation, she might well have decided not to undergo a mastectomy. However – and here we enter the twilight zone – what if that other woman had discovered she carried not only the dangerous BRCA1 mutation, but another mutation in the (fictional) gene ABCD3, which impairs a brain area responsible for evaluating probabilities, thereby causing people to underestimate dangers? What if a statistician pointed out to this woman that her mother, grandmother and several other relatives all died young because they underestimated various health risks and failed to take precautionary measures?

In all likelihood, you too will make important decisions about your health in the same way as Angelina Jolie. You will do a genetic test, a blood test or an fMRI; an algorithm will analyse your results on the basis of enormous statistical databases; and you will then accept the algorithm's recommendation. This is not